

The Transformer Architecture

Self-Attention · Multi-Head Attention · Positional Encoding · Encoder–Decoder

Outline

Attention

Multi-head attention

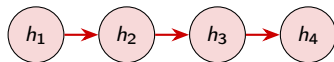
Positional encoding

The Transformer block

Why it works & practice

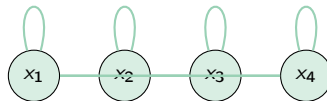
The key idea

LSTM



Sequential: $O(n)$ path

Transformer



Parallel: $O(1)$ path

“Attention Is All You Need” (Vaswani et al., 2017)

Parallel

Process all tokens
simultaneously

Direct access

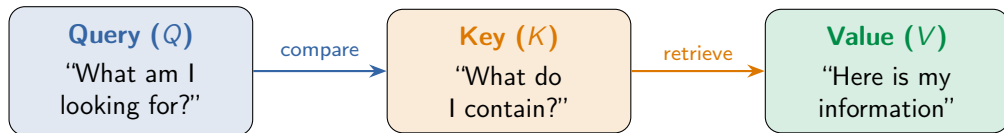
Any token can
attend to any other

Contextual

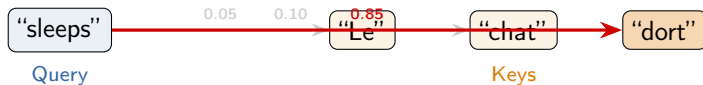
Representations
change with context

Attention — intuition

Think of it as a soft dictionary lookup

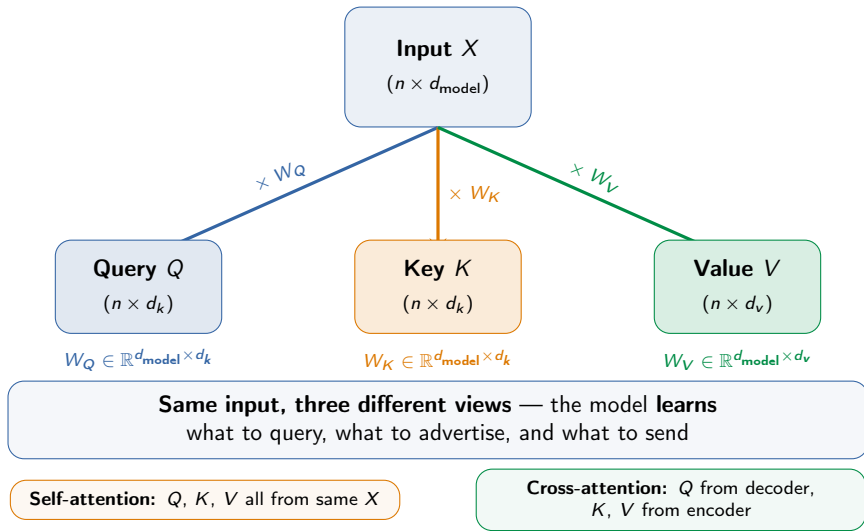


Example: translating "Le chat dort" → "The cat sleeps"



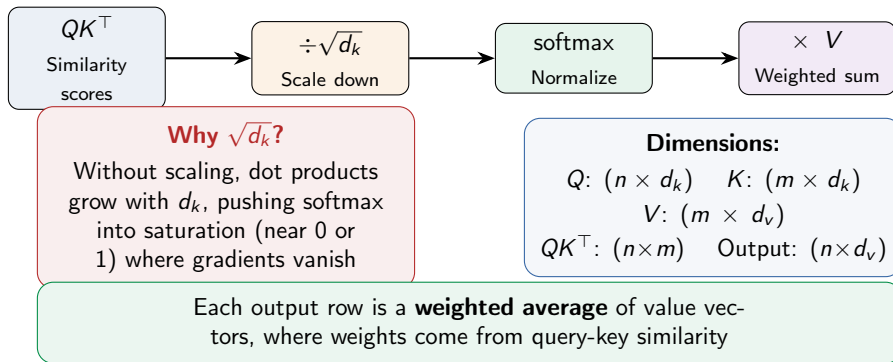
Output = weighted sum of values: $0.05 \cdot v_{\text{Le}} + 0.10 \cdot v_{\text{chat}} + 0.85 \cdot v_{\text{dort}}$

Q, K, V — where do they come from?



Scaled dot-product attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V$$



Attention — worked example

Tokens:

The

cat

sat

	$QK^T/\sqrt{d_k}$				Weights		
	The	cat	sat				
The	1.2	0.5	0.3		.49	.28	.23
cat	0.4	2.1	0.8	softmax →	.12	.63	.25
sat	0.2	0.9	1.5		.15	.30	.55

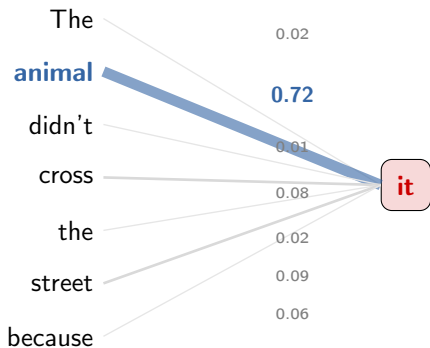
Row for “cat”: attends 63% to itself, 25% to “sat”, 12% to “The”

Output for “cat” =
 $0.12 \cdot v_{\text{The}} + 0.63 \cdot v_{\text{cat}} + 0.25 \cdot v_{\text{sat}}$

Each output is a **context-aware** representation — unlike Word2Vec, the same word gets different embeddings in different contexts

Attention learns meaningful relationships

“The **animal** didn’t cross the street because **it** was too tired”



Coreference resolved!

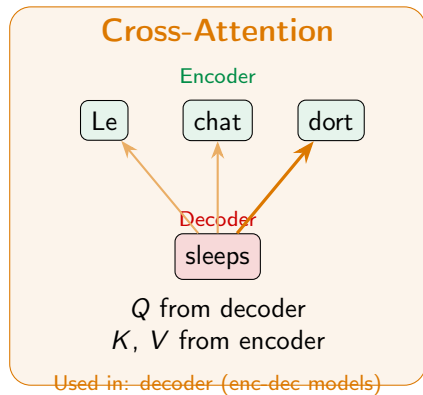
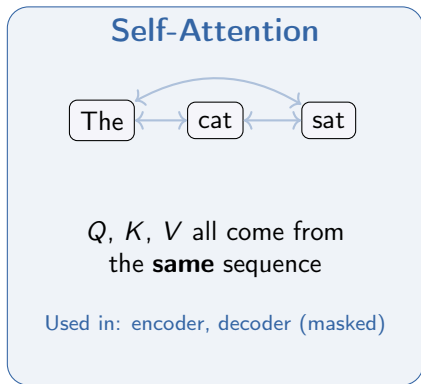
“it” attends most strongly to “animal” — the model learned that “it” refers to “animal”

No explicit rule

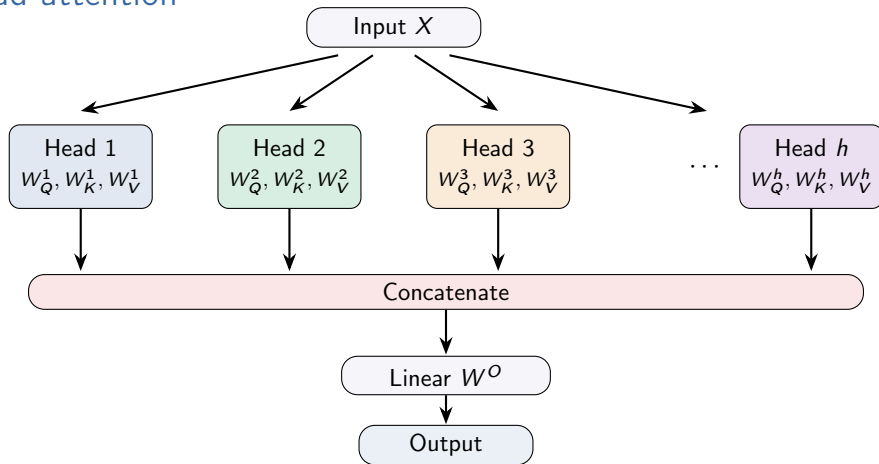
This emerges from training — the model learns which tokens are relevant for each query

Different attention heads specialize in different relationships (see next slides)

Self-attention vs. cross-attention



Multi-head attention

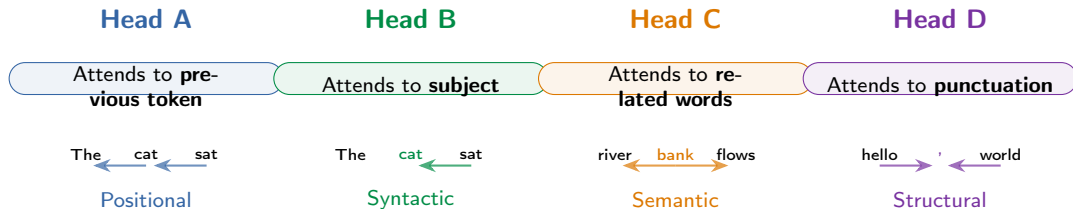


$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

where $\text{head}_i = \text{Attention}(QW_Q^i, KW_K^i, VW_V^i)$

What different heads learn

Different heads specialize in different types of relationships:

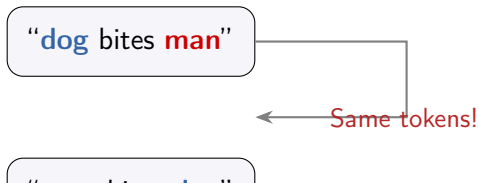


Multiple heads = multiple “perspectives” on the same input.

Typical: $h = 8$ (BERT-base) or $h = 12-96$ (larger models). $d_k = d_{\text{model}}/h$

Total compute is the same as single-head attention with full d_{model} , since each head uses $d_k = d_{\text{model}}/h$

Positional encoding — why we need it



Problem: Self-attention computes dot products between token pairs.

It's **permutation-invariant** — swapping token order doesn't change the output!

But word order matters: these two sentences mean very different things.

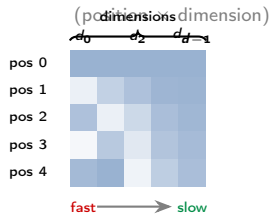
Solution: Add **positional information** to the input embeddings

$$\text{input}_i = \text{token_embedding}_i + \text{position_encoding}_i$$

Sinusoidal positional encoding

$$\begin{aligned} \text{PE}(\text{pos}, 2i) &= \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right) \\ \text{PE}(\text{pos}, 2i + 1) &= \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right) \end{aligned}$$

PE matrix



Unique pattern

Each position gets
a distinct encoding

No learned params

Fixed, deterministic,
works for any length

Relative positions

$\text{PE}_{\text{pos}+k}$ can be
expressed as linear
function of PE_{pos}

Multi-scale

Low dims = fine
position, high
dims = coarse

$$\text{input}_i = \text{embedding}(x_i) + \text{PE}(i) \quad (\text{element-wise addition})$$

Modern positional encodings

Sinusoidal

Vaswani et al., 2017

Added to embeddings

Absolute position

Fixed (not learned)

Used by: original Transformer

RoPE

Su et al., 2021

Rotates Q
and K vectors

Relative position

No extra parameters

Used by: LLaMA, Mistral

ALiBi

Press et al., 2022

Bias in attention scores

Linear distance penalty

No extra parameters

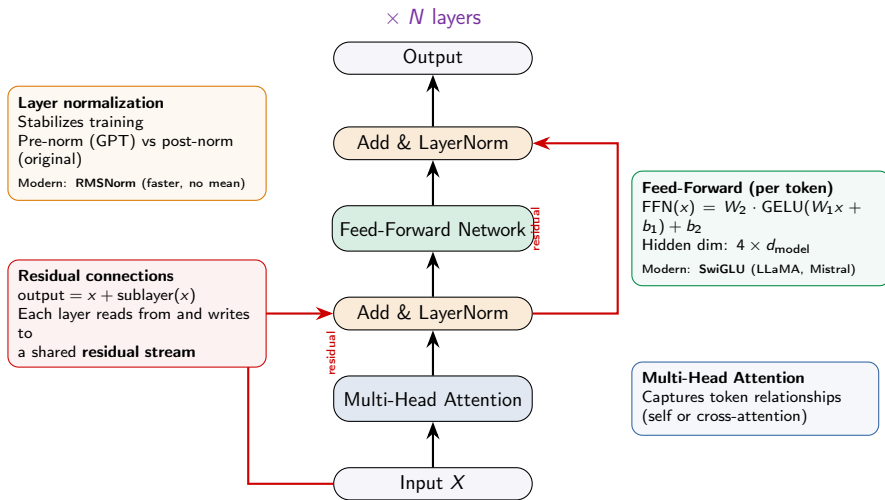
Used by: BLOOM, MPT

Where position info is injected:



RoPE is the most popular today — it naturally encodes relative position and supports length extrapolation with techniques like YaRN

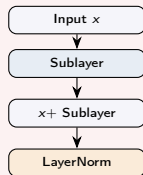
The Transformer block



Modern Transformer block variants

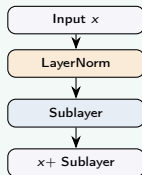
Three key changes in modern Transformer

Post-LN (original)



Unstable at large scale

Pre-LN (GPT, LLaMA)



Stable, no warmup needed

RMSNorm (Zhang & Sennrich, 2019)

$$\text{RMSNorm}(x) = \frac{x}{\text{RMS}(x)} \cdot \gamma$$

$$\text{RMS}(x) = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2}$$

Drops the mean-centering of Layer-Norm.

Faster: saves one reduction on per

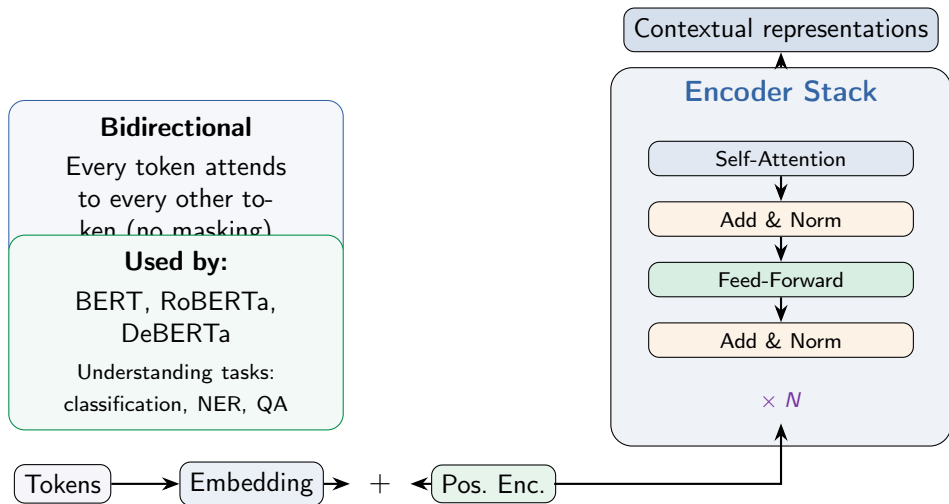
SwiGLU activation (Shazeer, 2020)

$$\text{SwiGLU}(x) = \text{Swish}(xW_1) \odot (xW_3)$$

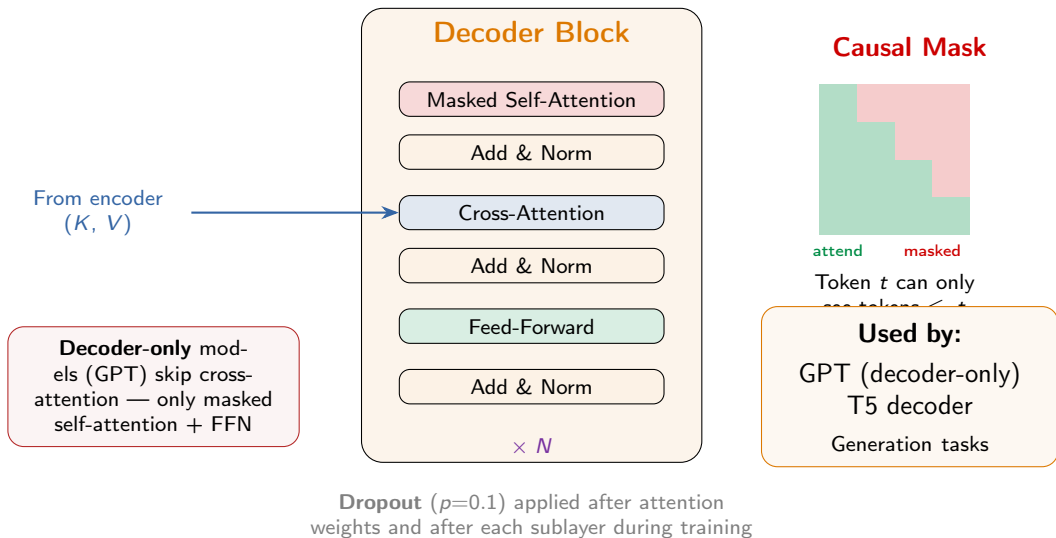
Replaces ReLU/GELU in FFN. Uses a **gating** mechanism (element-wise product of two projections). Third weight matrix W_3 added; hidden dim adjusted to $\frac{8}{3}d$ to match param count.

Modern recipe: **Pre-LN + RMSNorm + SwiGLU + RoPE** — small changes, large impact on training stability and performance

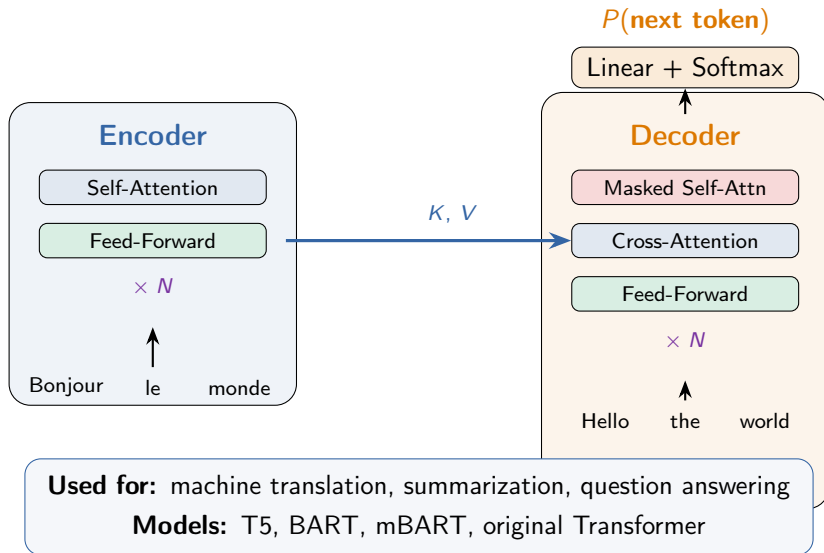
The Transformer encoder



The Transformer decoder



Encoder-Decoder architecture



Three Transformer architectures

Encoder-Only

Bidirectional
MLM training
Understanding tasks

BERT, RoBERTa

Decoder-Only

Autoregressive ($L \rightarrow R$)
CLM training
Generation tasks

GPT, LLaMA, Mistral

Encoder-Decoder

Bidirectional enc. +
autoregressive dec.
Seq-to-seq tasks

T5, BART, mBART

Enc-Only

Dec-Only

Enc-Dec

Direction	Bidirectional	Left-to-right	Both
Best for	Classification, NER	Text generation	Translation, summary
Today's trend	Less common	Dominant	Niche

The trend since GPT-3 (2020): **decoder-only** models at scale
can do almost everything, including understanding tasks.

How Transformers learn

Causal LM (decoder)

"The cat sat on the ____"

Transformer

softmax over vocab

$$P(\text{"mat"}) = 0.42$$

$$\mathcal{L} = - \sum_{t=1}^T \log P(x_t \mid x_{<t})$$

Cross-entropy on **every** position

~~Teacher forcing: use true tokens
as input, not model's predictions~~

Masked LM (encoder)

"The **[MASK]** sat on the mat"

Transformer

softmax over vocab

$$P(\text{"cat"}) = 0.67$$

$$\mathcal{L} = - \sum_{i \in \mathcal{M}} \log P(x_i \mid x_{\setminus \mathcal{M}})$$

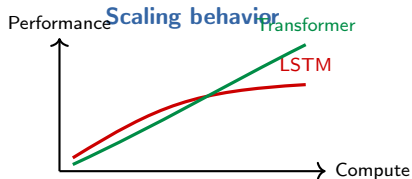
Loss only on **masked** positions

~~15% of tokens masked: 80% [MASK],
10% random, 10% unchanged~~

Since GPT-3: **causal LM is the dominant paradigm** —
one training objective, scales to any task via prompting

Why Transformers won

	LSTM	Transformer
Parallelizable	No (h_t needs h_{t-1})	Yes (all tokens at once)
Long-range path	$O(n)$ steps	$O(1)$ (direct attention)
Context	Limited by hidden size	Full context window
Scaling	Diminishing returns	Log-linear improvement
Training speed	Slow (sequential)	Fast (GPU-friendly)



The scaling insight:

Transformers reliably improve with more data, parameters, and compute

This enabled the LLM revolution:
GPT-3, PaLM, LLaMA, Claude, ...

Computational complexity

Self-Attention

$O(n^2 \cdot d)$ time

$O(n^2)$ memory

(QK^T is an $n \times n$ matrix)

Feed-Forward

$O(n \cdot d^2)$ time

$O(d^2)$ memory

(two $d \times 4d$ weight matrices)

Which dominates?

Short context ($n < d$)

FFN dominates

E.g., $n=512$, $d=4096$

Attention: $0.5M \cdot d$ vs FFN: $512 \cdot 16M$

Long context ($n > d$)

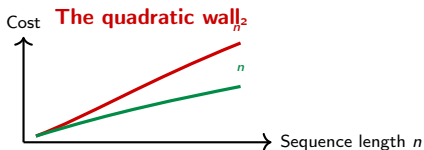
Attention dominates

E.g., $n=128K$, $d=4096$

Doubling sequence length:

- Attention: $4\times$ more compute
- KV cache memory: $2\times$ more
- Attention matrix: $4\times$ more memory

128K tokens $\rightarrow$ 16 **billion** entries per head per layer



This quadratic bottleneck motivates FlashAttention, sparse patterns, and state space models

Transformer dimensions in practice

Model	Layers	d_{model}	Heads	d_{ff}	Params
BERT-base	12	768	12	3072	110M
BERT-large	24	1024	16	4096	340M
GPT-2	12	768	12	3072	117M
GPT-3	96	12288	96	49152	175B
LLaMA-2 7B	32	4096	32	11008	7B
LLaMA-2 70B	80	8192	64	28672	70B

$$d_k = \frac{d_{\text{model}}}{h}$$

Head dimension

$$d_{\text{ff}} \approx 4 \times d_{\text{model}}$$

FFN hidden size

$$\text{Params} \approx 12 \cdot N \cdot d^2$$

Rough estimate

Further reading

Attention & Transformers

- Vaswani et al. (2017), “Attention Is All You Need” — the original Transformer paper
- Bahdanau et al. (2015), “Neural Machine Translation by Jointly Learning to Align and Translate”
- Jay Alammar, “The Illustrated Transformer” —

Positional Encodings

- Su et al. (2021), “RoFormer: Enhanced Transformer with Rotary Position Embedding” (RoPE)

Architecture Variants & Surveys

- Devlin et al. (2019), “BERT: Pre-training of Deep Bidirectional Transformers” (encoder-only)
- Radford et al. (2018/2019), “Improving/Language Models are Unsupervised Multitask Learners” (GPT/GPT-2)
- Lin et al. (2022), “A Survey of Transformers” — comprehensive taxonomy

Questions?

Next: Tokenization